
Introduction to Java

Core Applications

Agenda

01 Core Java

- Introduction
- Variables, Data Types, and Operators
- Control Flow
- Java I/O
- Strings
- Arrays
- Access Modifiers
- Exception Handling
- Multithreading

What is Java ?

Java is a high-level, general-purpose programming language known for its portability, versatility, and security features. Developed by Sun Microsystems, Java was released in 1995 and has since become one of the most popular programming languages in the world.

Java is a widely-used programming language for coding software applications. It has been a popular choice among developers for over two decades, with millions of Java applications in use today. Java is a multi-platform, object-oriented, and network-centric language that can be used as a platform in itself. It is a fast, secure, reliable programming language for coding everything from mobile apps and enterprise software to big data applications and server-side technologies.



History

Java is a general purpose, Object oriented programming Language developed by **Sun Microsystems of USA in 1991**. Originally called ***Oak*** by James Gosling.

Java Milestones



✓ **1990** – Sun Microsystems decided to develop special software for consumer electronic devices headed by James Gosling.



✓ **1991** – Using C++ the team announced a new language named ***Oak***.



✓ **1992** – The Green project team by Sun, shown new language to control home appliances using hand held device with tiny touch-sensitive screen.



✓ **1993** - The www appeared on the internet and transformed the text based internet into graphical rich environment using Web Applets.



✓ **1994** – Team developed a web browser called “HotJava” for web Applets.



✓ **1995** – “Oak” was renamed “Java” due to some legal snags. Java is just a name. Netscape and Microsoft supports to Java.

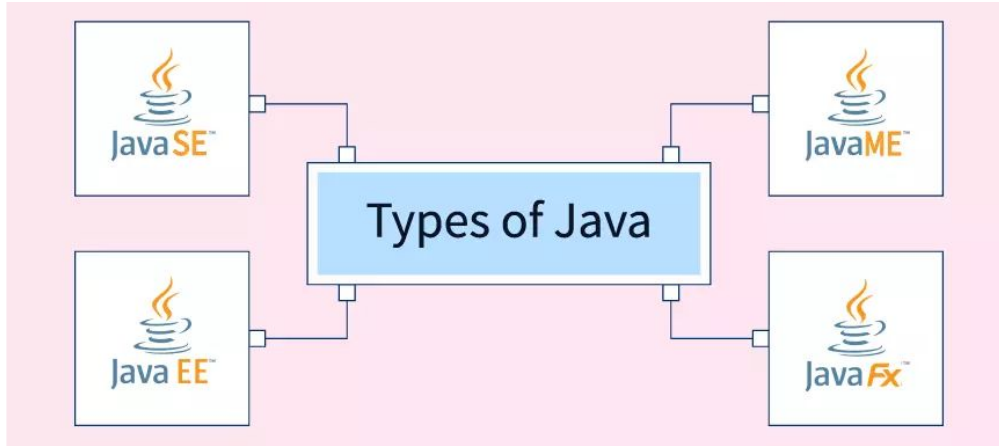
✓ **1996** – Java established itself as a leader for internet & general purpose programming. Java founds its home.

Why Java ?

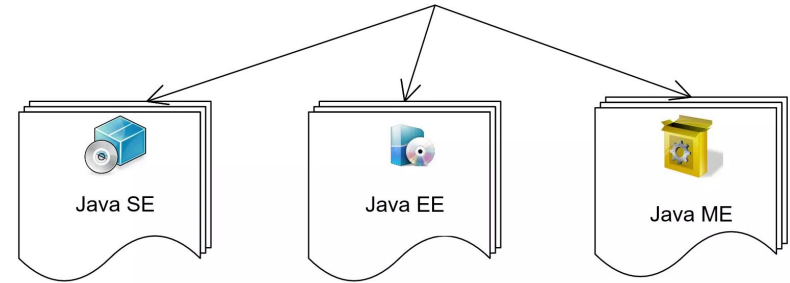


Development Environment

From Java 1.2 to Java 11



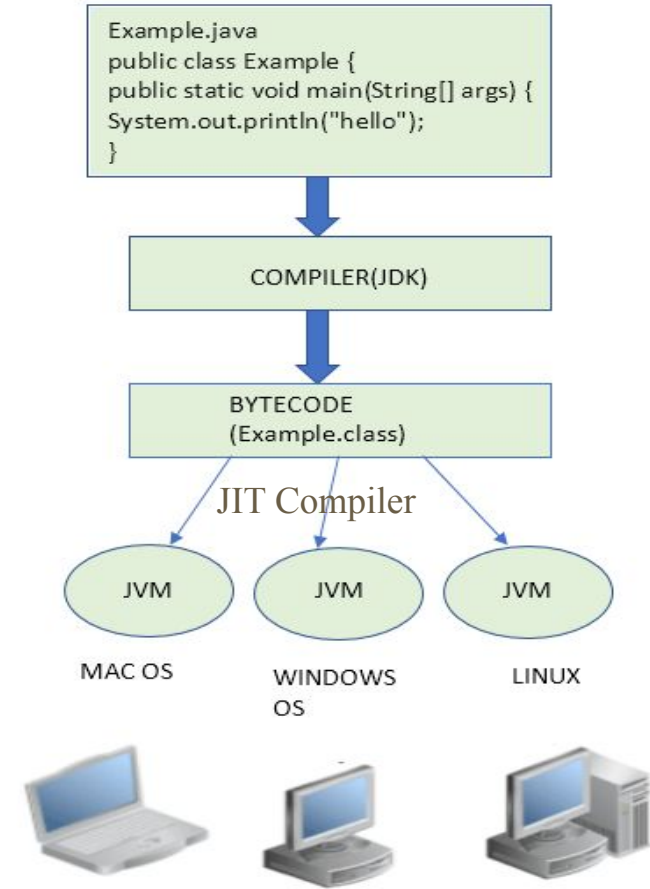
After Java 11



Development Environment

Java SE	https://www.oracle.com/java/technologies/downloads/
IDE	Notepad/Notepad++/eclipse/Netbeans etc.,
Notepad++	https://notepad-plus-plus.org/downloads/
Eclipse	https://www.eclipse.org/downloads/

Execution Flow

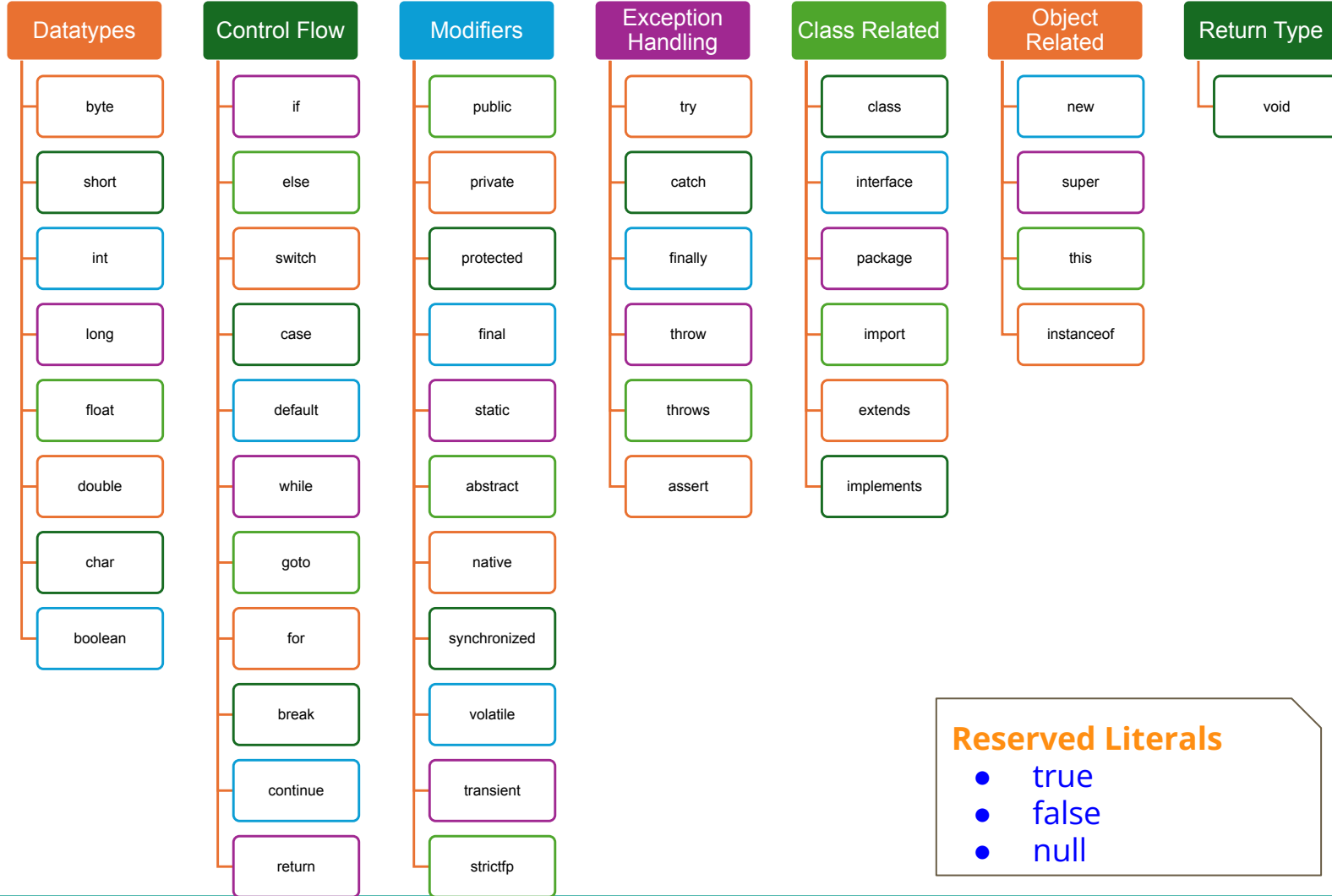


Applications

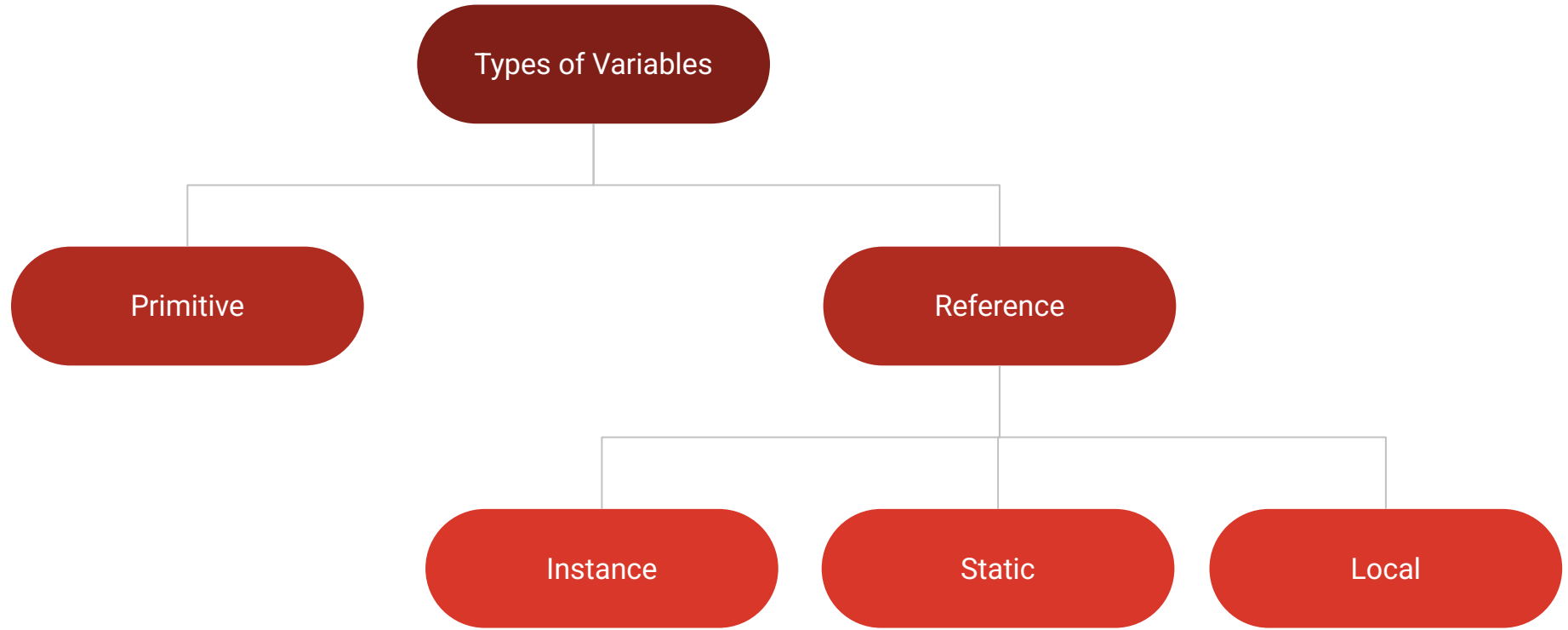
1. Console
2. Point of Sale
3. Web
4. Networking
5. Mobile
6. Game Development
7. Cloud Computing
8. Big Data
9. Artificial Intelligence
10. Internet of Things



Reserved Keywords

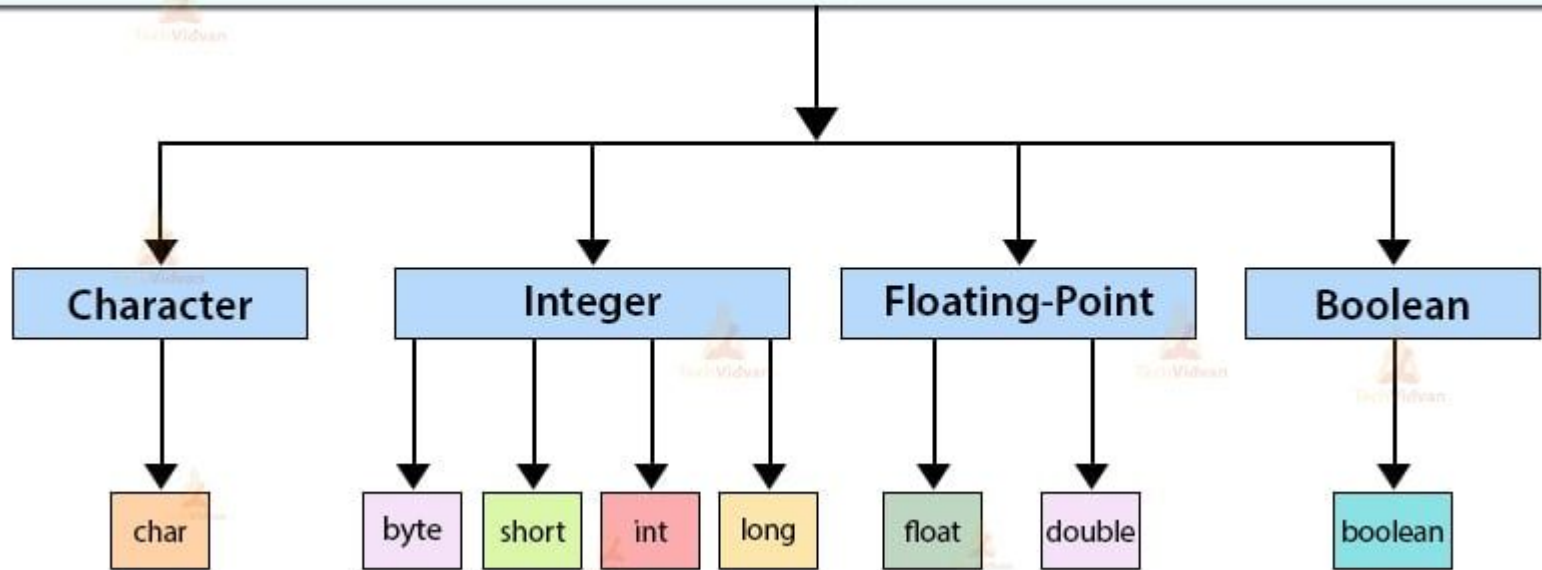


Variables

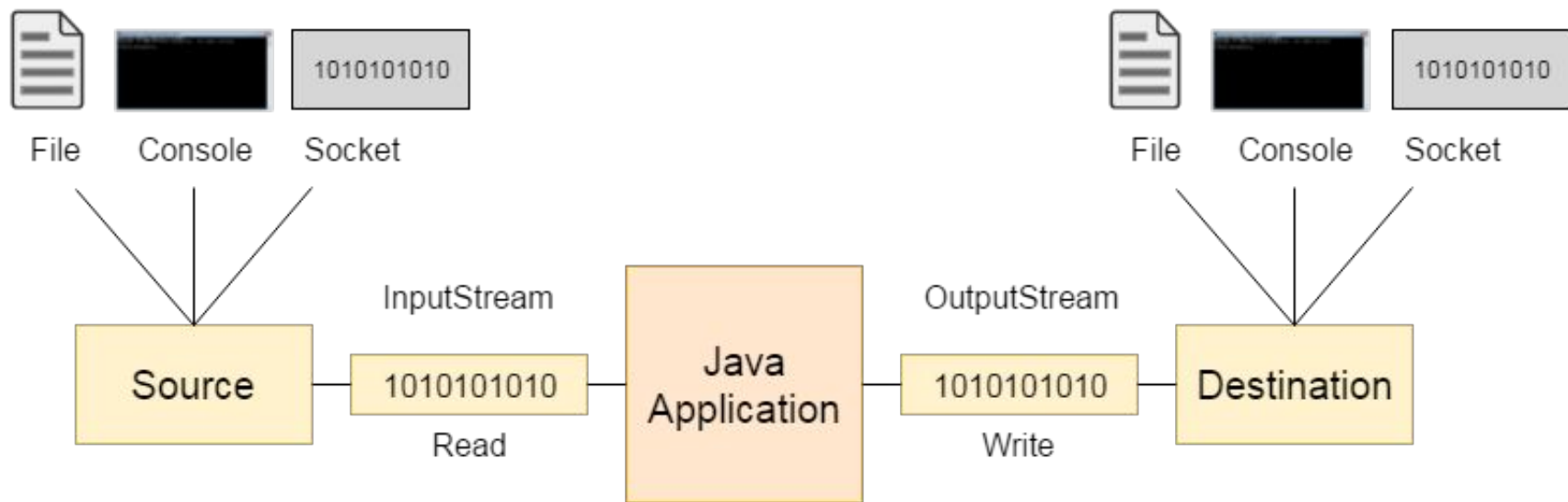


Data Types

Primitive Data Types in Java



I/O



```
package edu.training.byts;
```

Package declaration

```
import java.lang.*;
```

Import statements

```
/**  
 * @author Sairam Sekhar P  
 */
```

Comments

```
interface Test {  
    public void display();  
}
```

Interface statements

```
public class Structure {  
    int a = 10;
```

Class name

Instance variable

```
    public static void main(String[] ars) {  
        System.out.print("Hello Java!");  
    }
```

Main method

```
    public void display() {  
        String name = "Samantha";  
        System.out.println("Name: " + name);  
        System.out.println("Value in a is: " + a);  
    }
```

=> Method

Class Definition

```
}
```

Java Structure

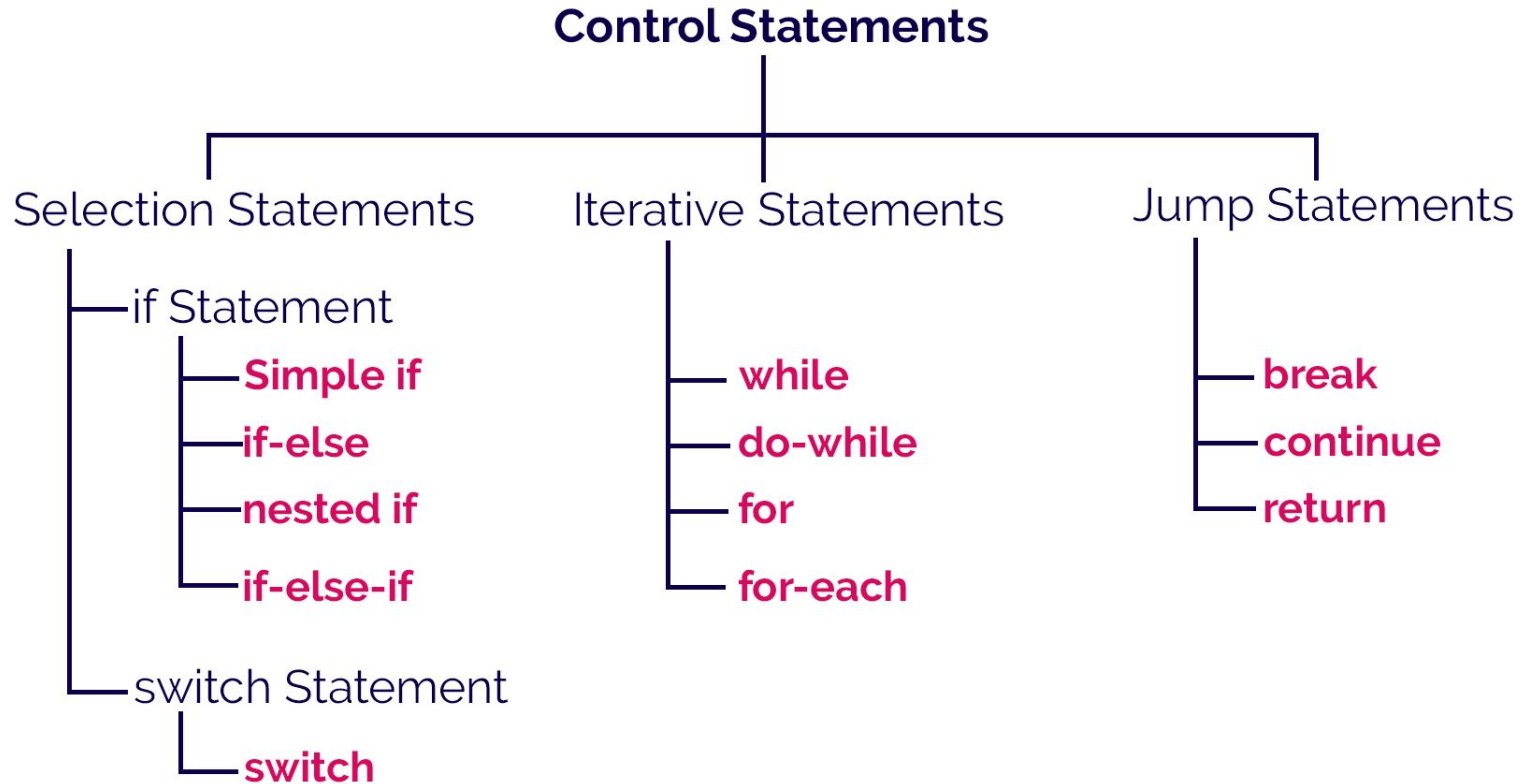
Operators



Reserved Operators

- `()` => type casting
- `instanceof`
- `new`
- `[]`

Control Statements



Arrays

Types of Arrays

Single Dimensional Array

array variable name
element type
length

```
int[] myInts = new int[10];  
int[] intValues = {1, 5, 18, 3, 7, 9, 23, 5, 11, 2};
```

Type of array
array variable name
List of Initialization Values

Multidimensional Array

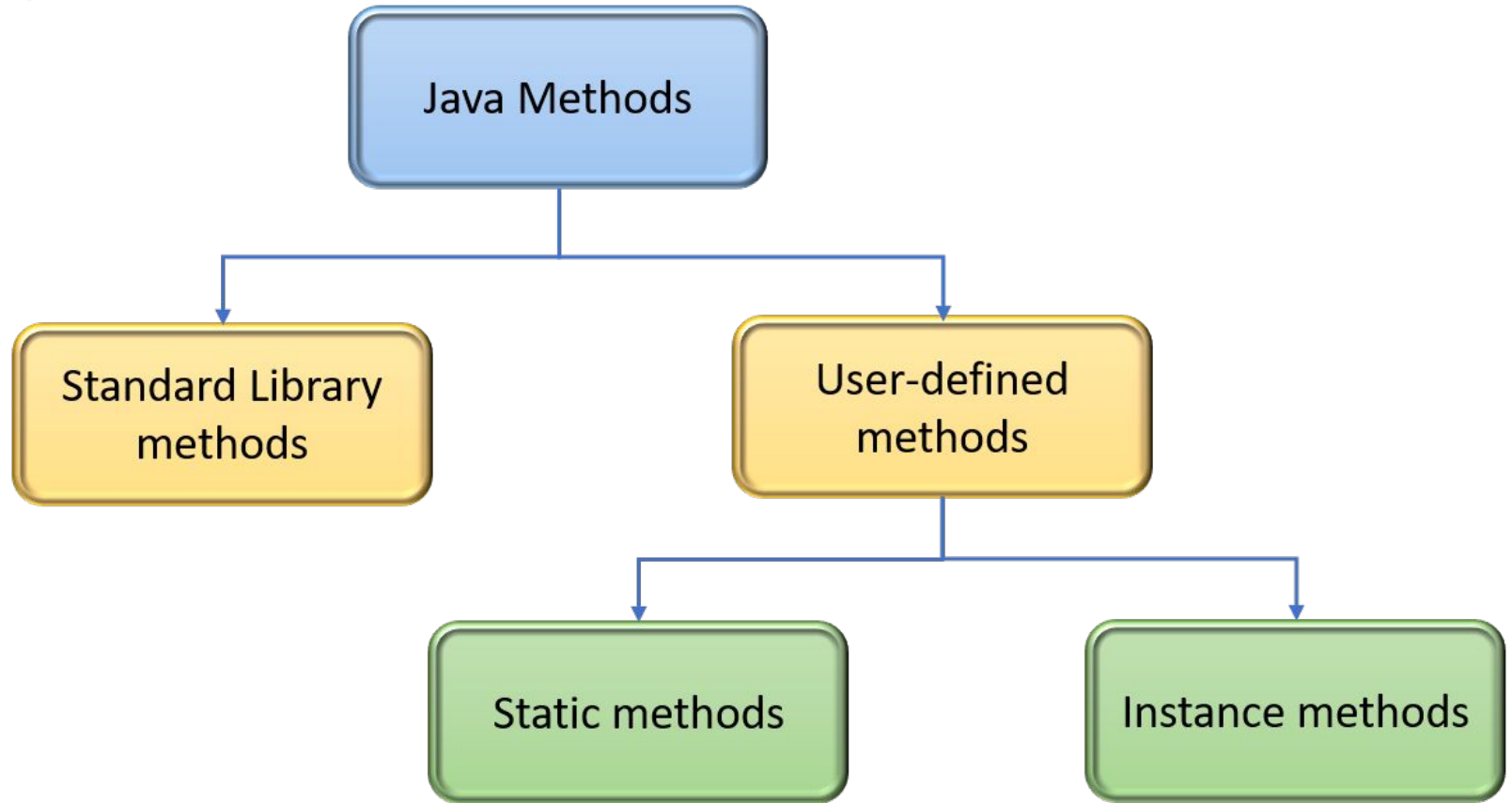
2D-Array

	Column 0	Column 1	Column 2
Row 0	X[0][0]	X[0][1]	X[0][2]
Row 1	X[1][0]	X[1][1]	X[1][2]
Row 2	X[2][0]	X[2][1]	X[2][2]

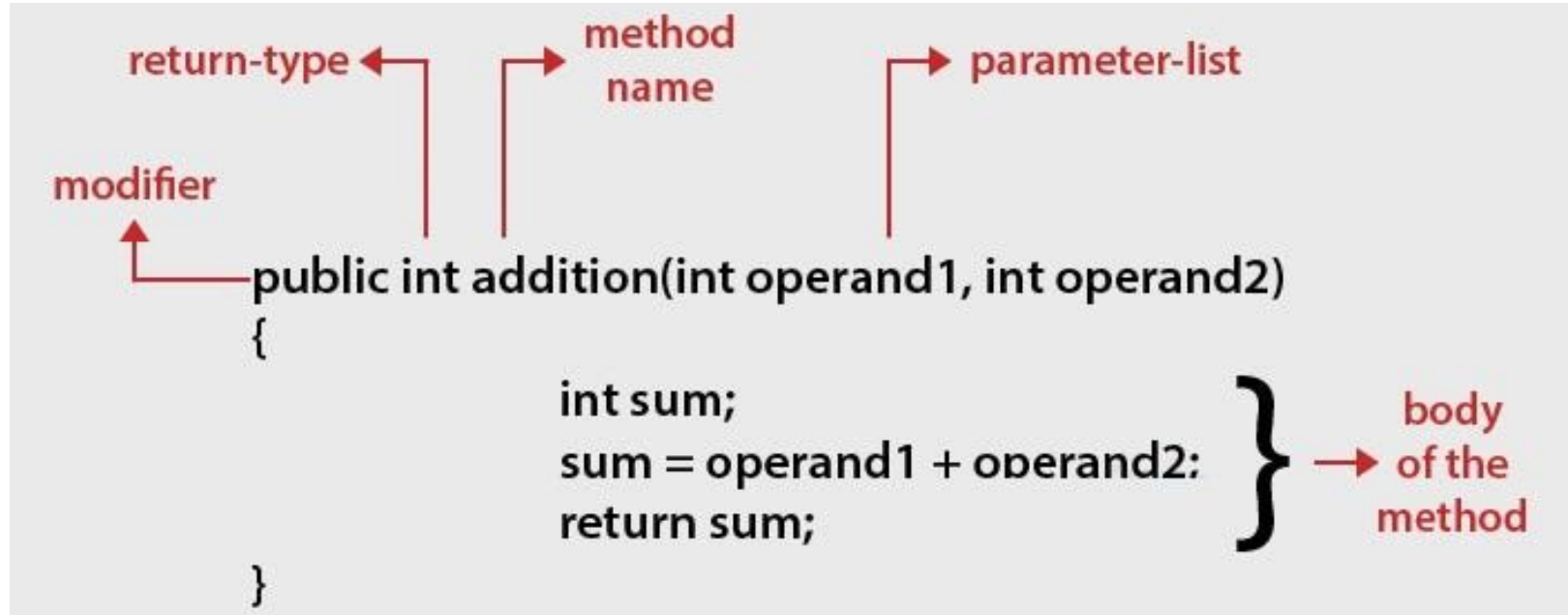
X Dimensional
Y Dimensional
Z Dimensional

3D-Array

Methods



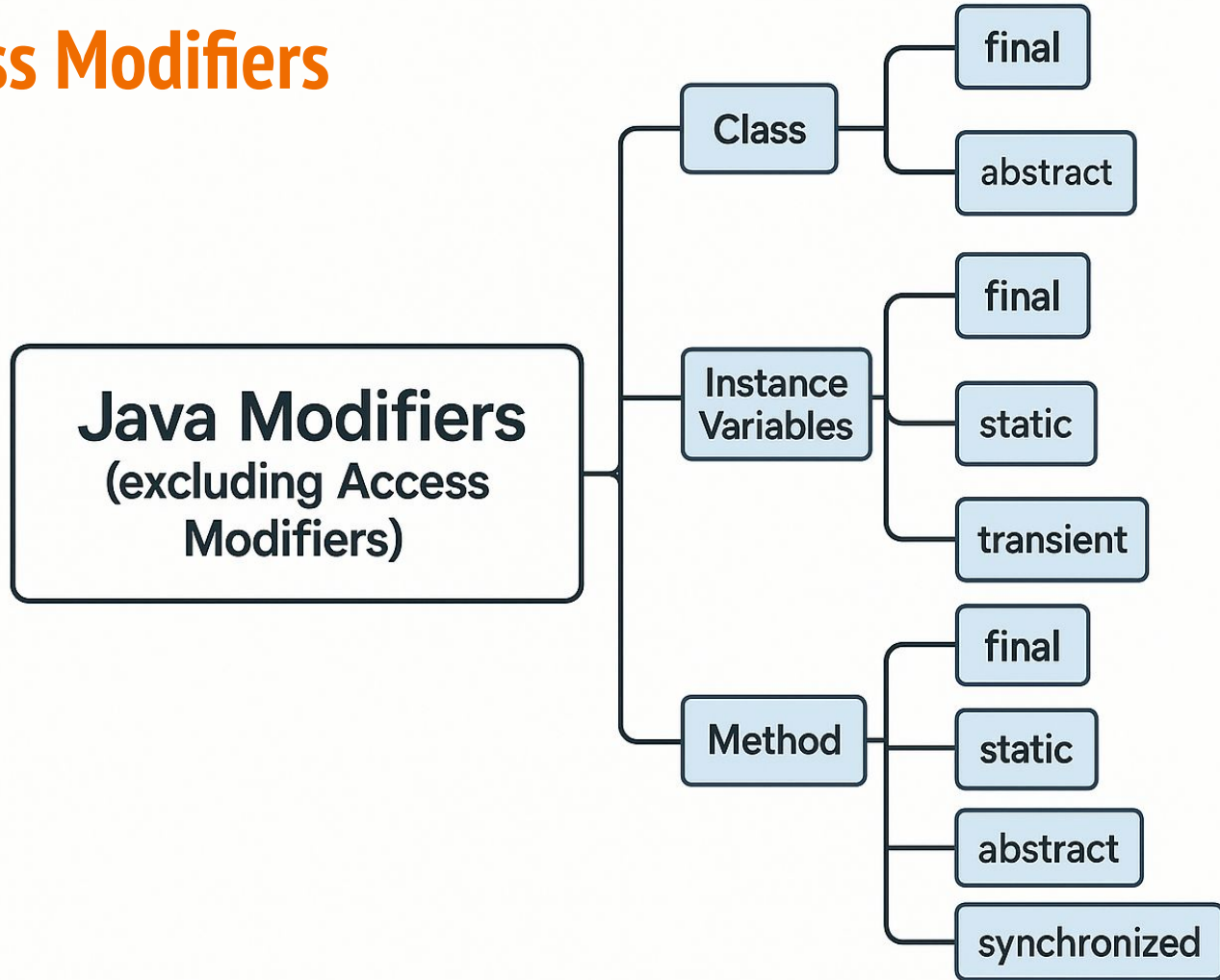
Methods



Access Modifiers

		public	private	protected	default
Same Package	Class	YES	YES	YES	YES
	Sub class	YES	NO	YES	YES
	Non sub class	YES	NO	YES	YES
Different Package	Sub class	YES	NO	YES	NO
	Non sub class	YES	NO	NO	NO

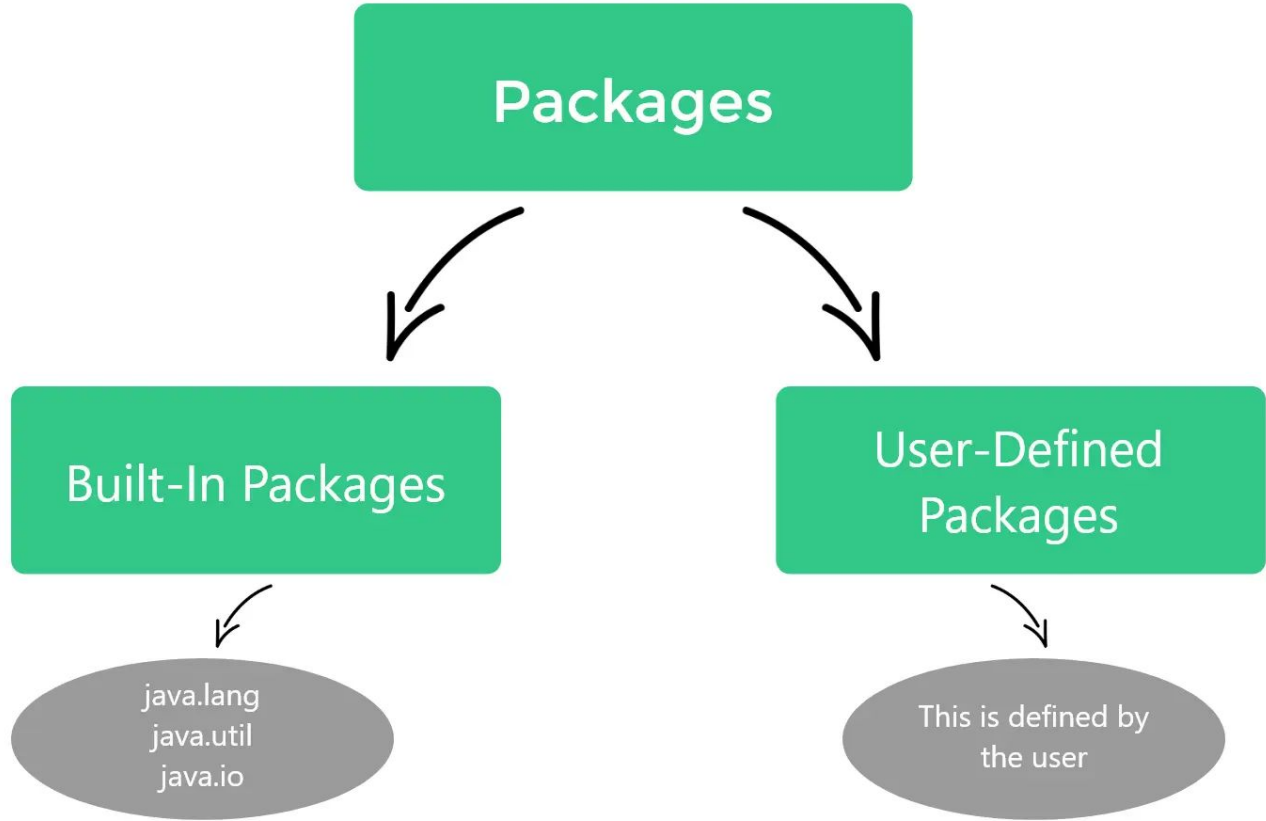
Non-Access Modifiers



Java Modifiers

Modifier	Class	Variable	Method	Meaning
public	✓	✓	✓	Accessible everywhere
private	✗	✓	✓	Same class only
protected	✗	✓	✓	Same package + subclass
default	✓	✓	✓	Same package
static	✗	✓	✓	Belongs to class
final	✓	✓	✓	No override/change/subclass
abstract	✓	✗	✓	Must be implemented
synchronized	✗	✗	✓	Thread-safe
volatile	✗	✓	✗	Visible to all threads
transient	✗	✓	✗	Not serialized
strictfp	✓	✗	✓	Consistent float calculations
native	✗	✗	↓ ✓	Implemented in other languages

Packages



Custom Package



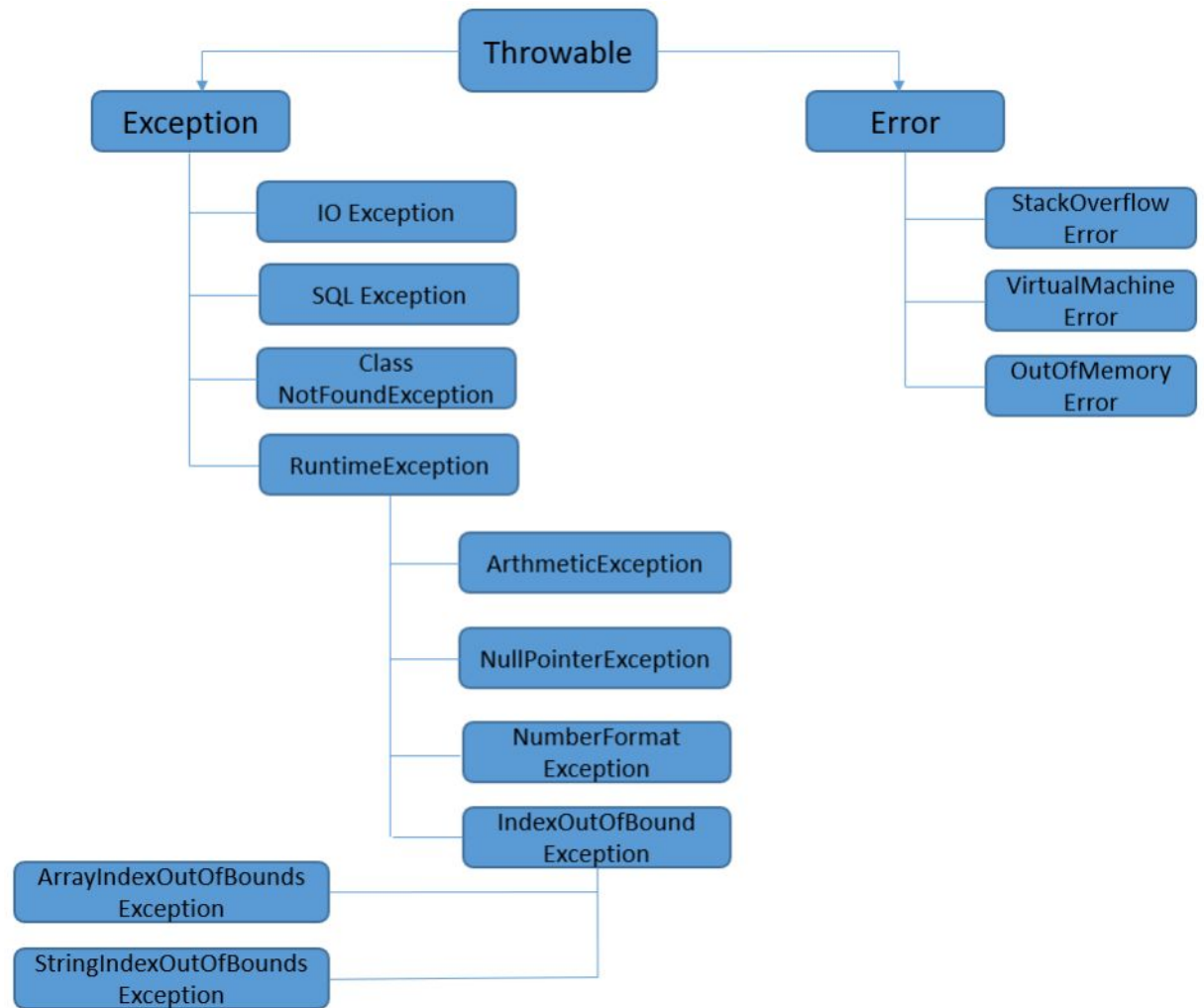
Exception Handling

```
try
{
    statements
}

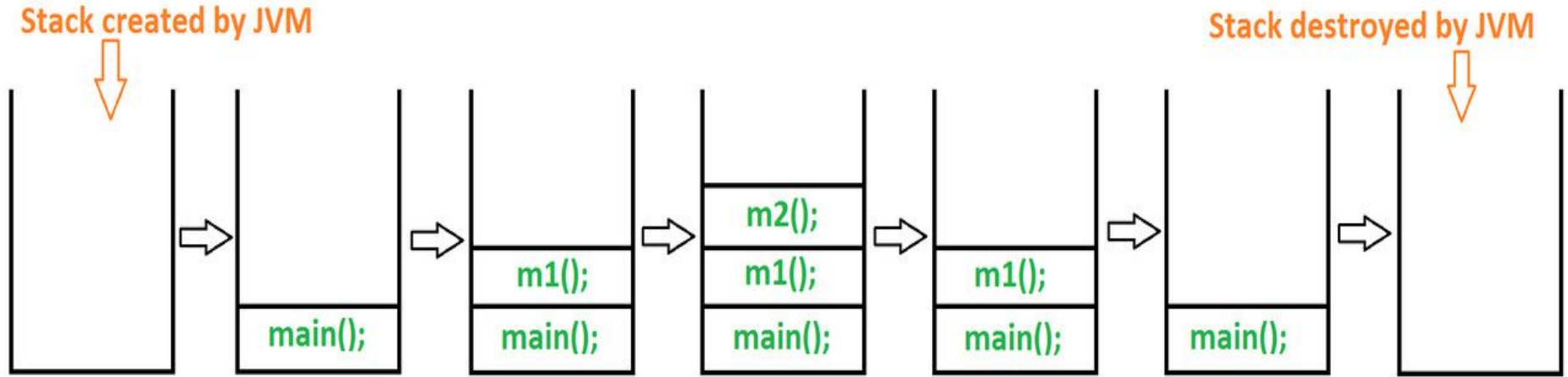
catch (Exception class object)
{
    statements
}

finally
{
    //Close file handles
    //Free the resources
}
```

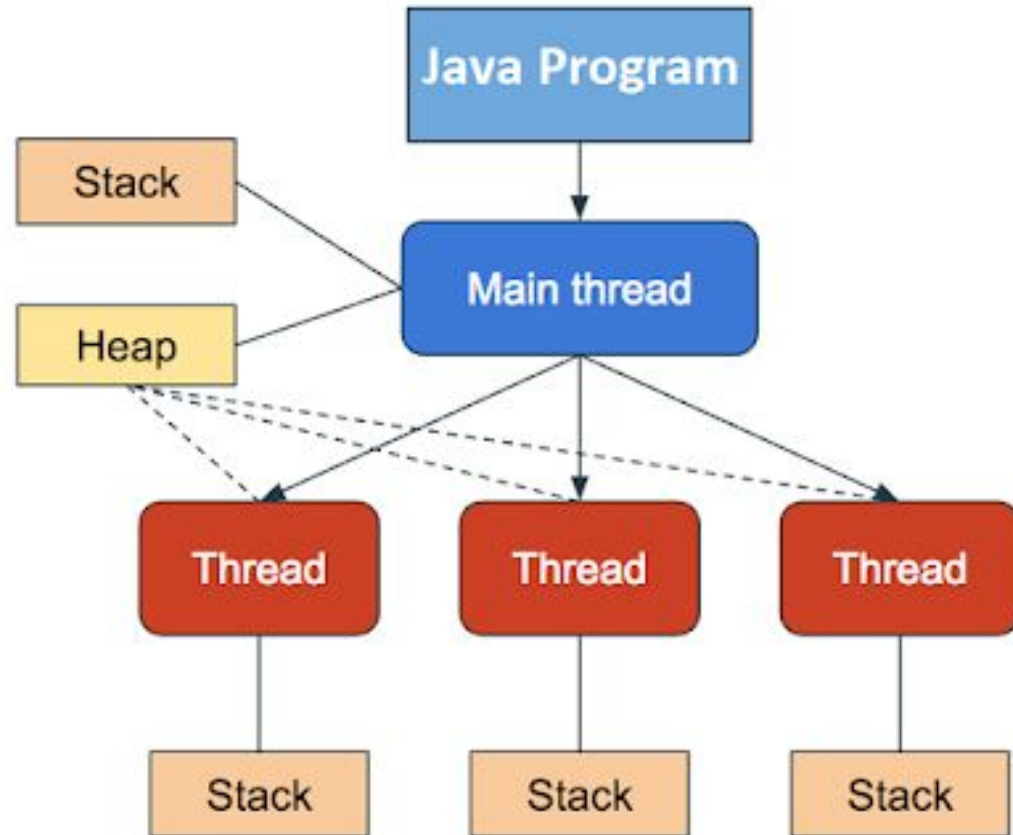
Exception Handling



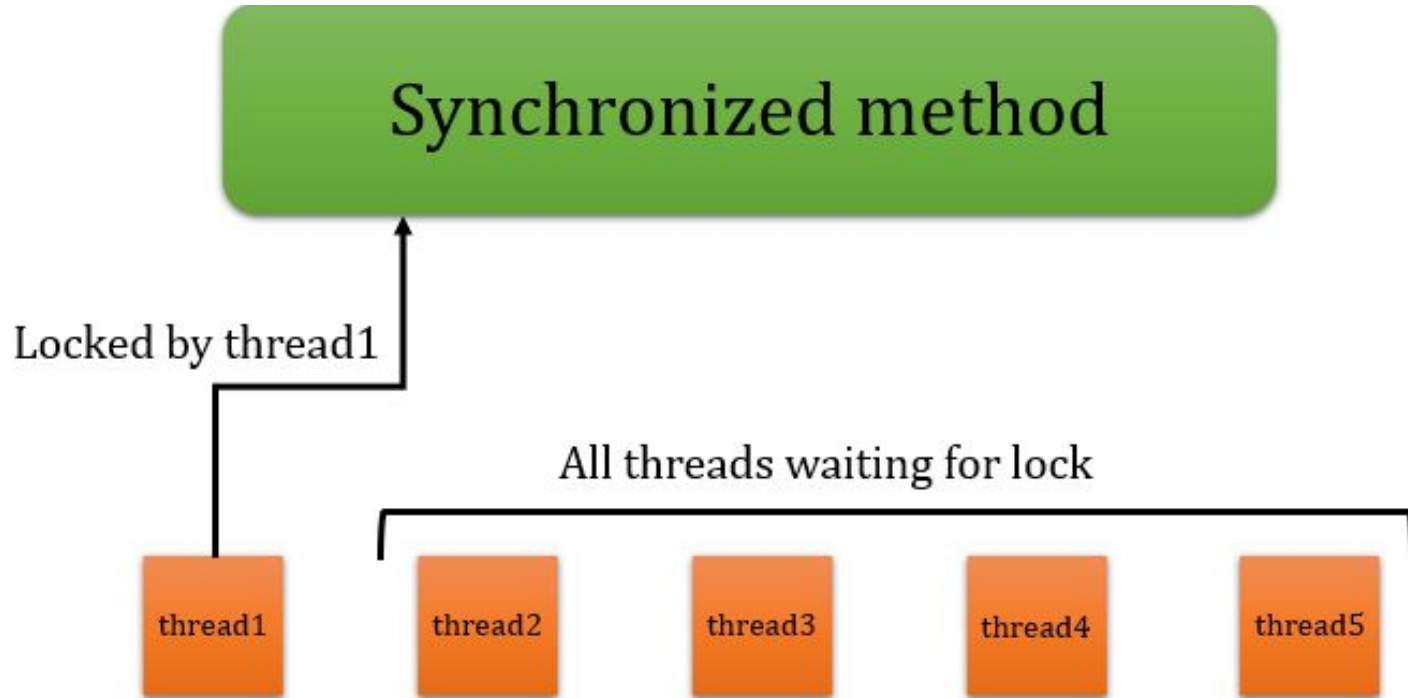
Java Runtime Stack Mechanism



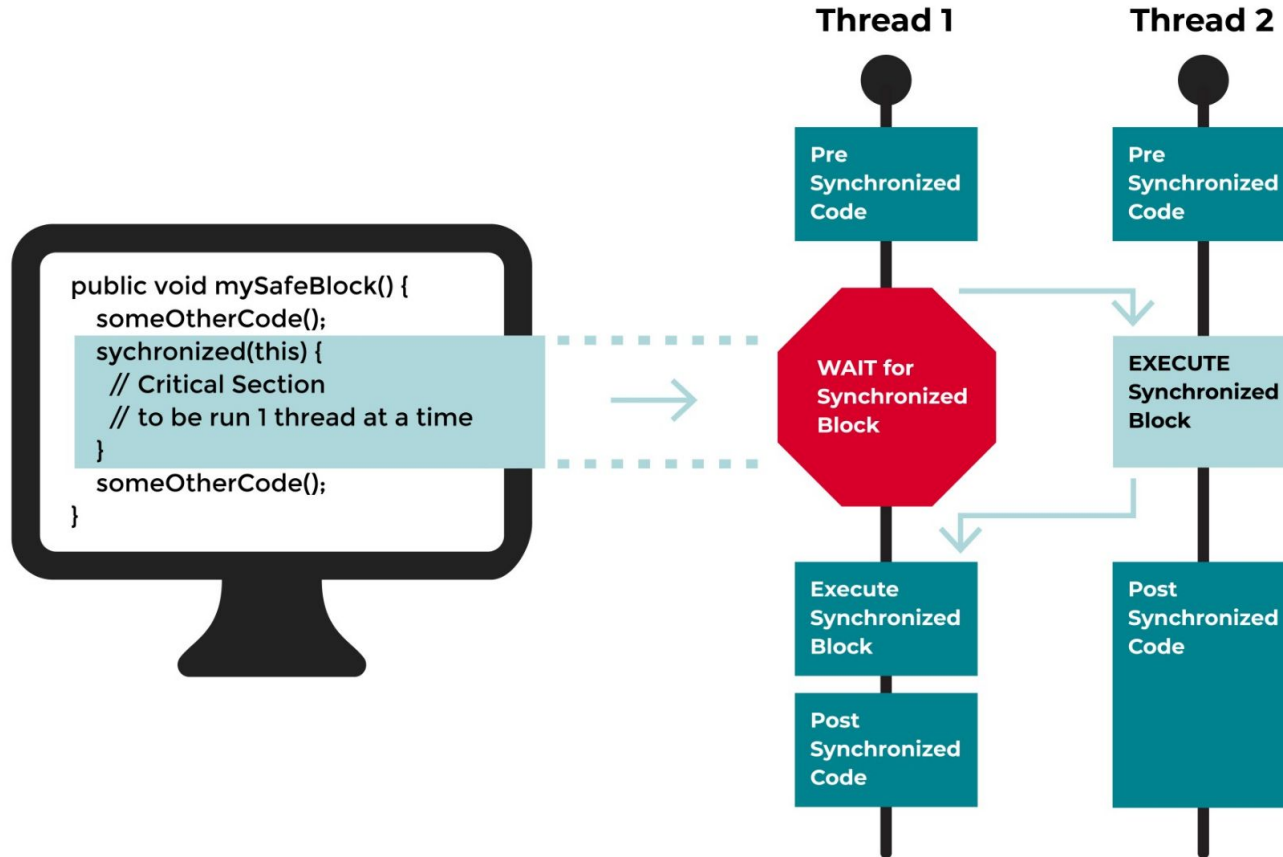
Multi Threading

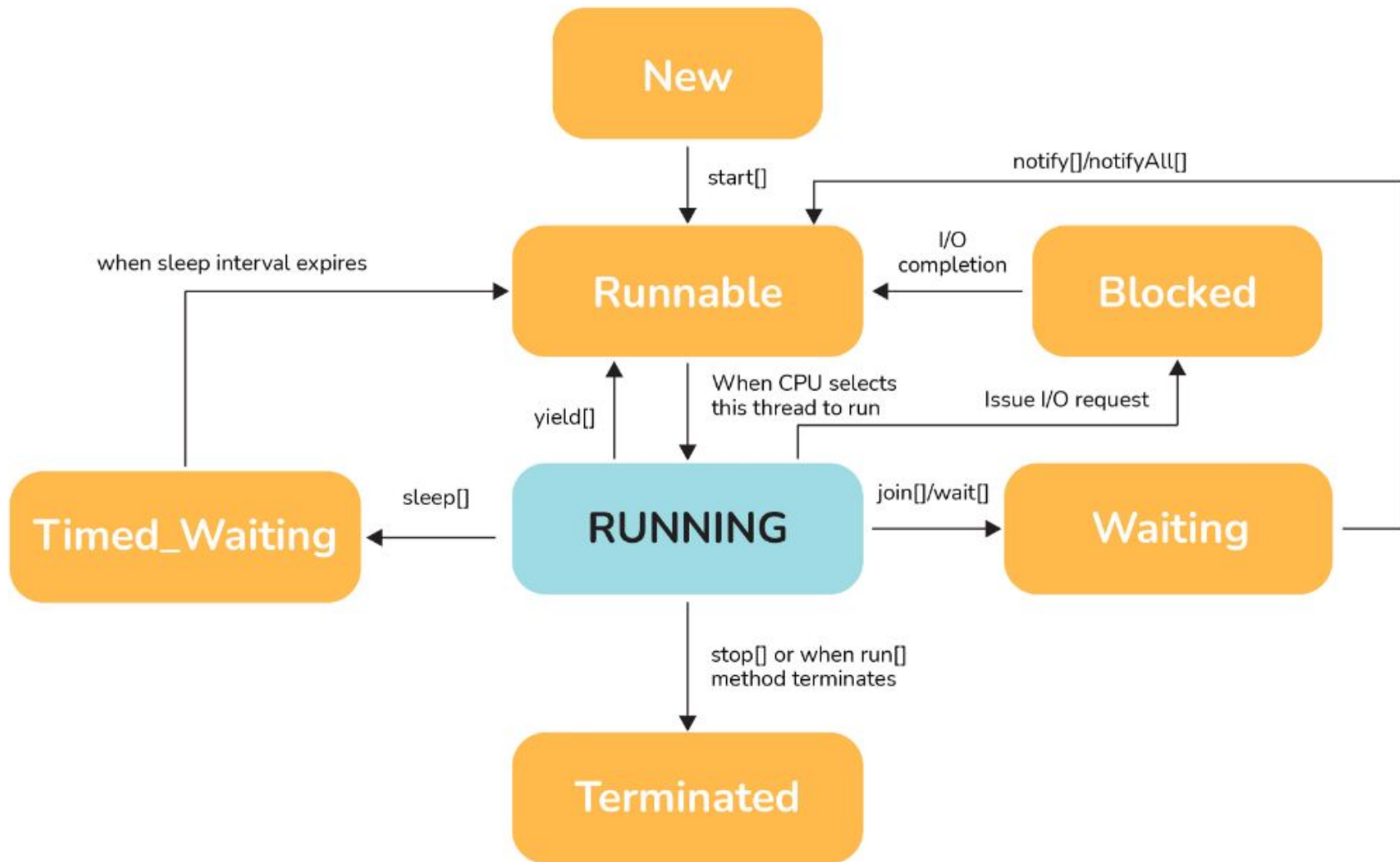


Synchronization

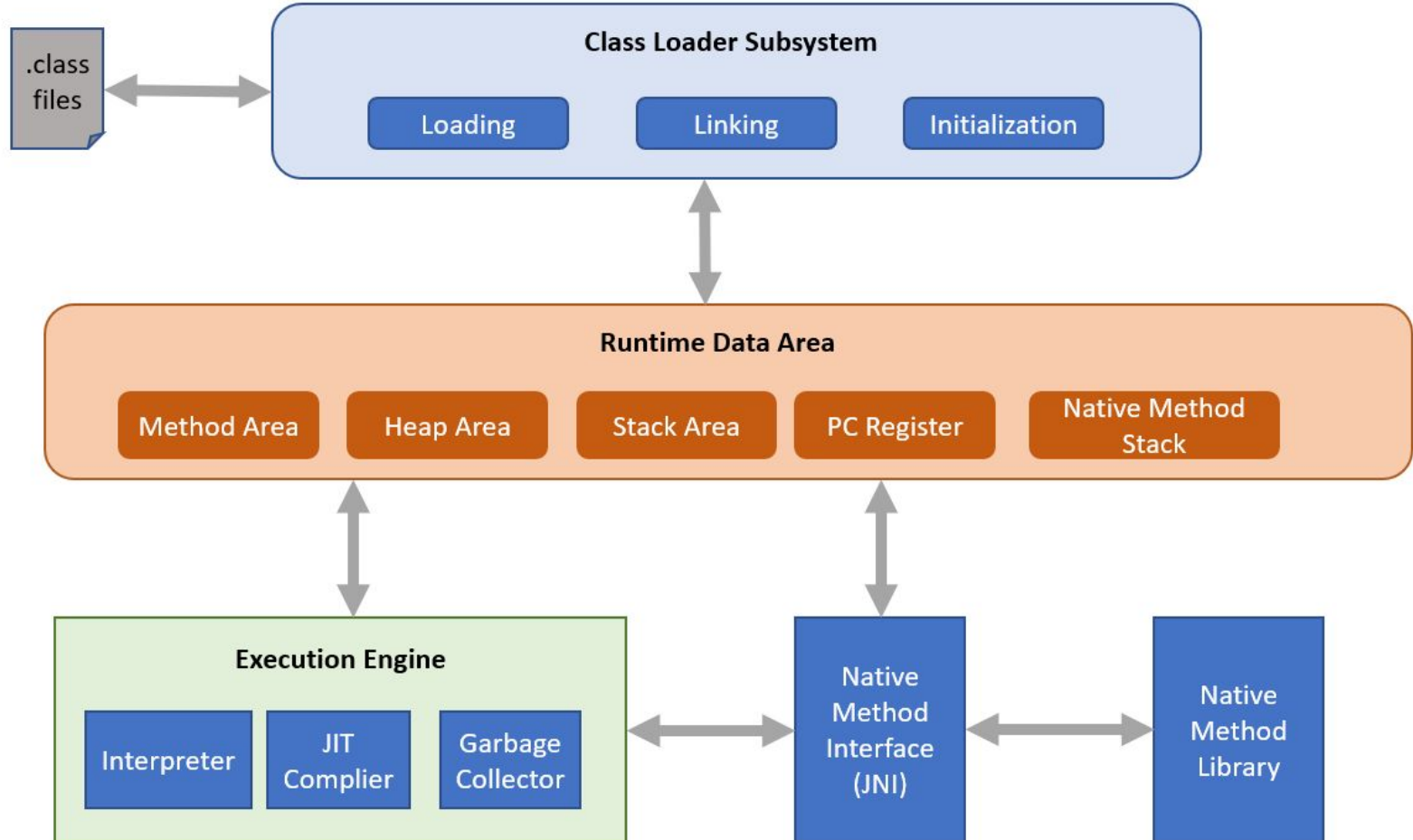


Synchronization





JVM Architecture



Strings

Factor / Class	String	StringBuffer	StringBuilder
Mutability	Immutable	Mutable	Mutable
Thread Safety	Not thread safe	Thread safe	Not thread safe
Performance	Very high	Moderate	Very high



Thank You

Accelerate your evolution – and stay a step ahead!